

▼ Figure 38.12 Exploring Fruit and Seed Dispersal

A plant's life depends on finding fertile ground. But a seed that falls and sprouts beneath the parent plant will stand little chance of competing successfully for nutrients. To prosper, seeds must be widely dispersed. Plants use biotic dispersal agents as well as abiotic agents such as water and wind.

Dispersal by Water

► Some buoyant seeds and fruits can survive months or years at sea. In coconut, the seed embryo and fleshy white “meat” (endosperm) are within a hard layer (endocarp) surrounded by a thick and buoyant fibrous husk.



Dispersal by Wind

► With a wingspan of 12 cm, the giant seed of the tropical Asian climbing gourd *Alsomitra macrocarpa* glides through the air of the rain forest in wide circles when released.



▼ The winged fruit of a maple spins like a helicopter blade, slowing descent and increasing the chance of being carried farther by horizontal winds.



► Tumbleweeds break off at the ground and tumble across the terrain, scattering their seeds.



▲ Some seeds and fruits are attached to umbrella-like “parachutes” that are made of intricately branched hairs and often produced in puffy clusters. These dandelion “seeds” (actually one-seeded fruits) are carried aloft by the slightest gust of wind.

Dispersal by Animals



◀ The sharp, tack-like spines on the fruits of puncture vine (*Tribulus terrestris*) can pierce bicycle tires and injure animals, including humans. When these painful “tacks” are removed and discarded, the seeds are dispersed.

► Seeds in edible fruits are often dispersed in feces, such as the black bear feces shown here. Such dispersal may carry seeds far from the parent plant.



◀ Some animals, such as squirrels, hoard seeds or fruits in underground caches. If the animal dies or forgets the cache's location, the buried seeds are well positioned to germinate.

► Ants are chemically attracted to seeds with “food bodies” rich in fatty acids, amino acids, and sugars. The ants carry the seed to their underground nest, where the food body (the lighter-colored portion shown here) is removed and fed to larvae. Due to the seed's size, unwieldy shape, or hard coating, the remainder is usually left intact in the nest, where it germinates.

